

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)		Substitution: power = $P = \frac{2.50}{10}$ (1) Power = 0.25 [kW] (1)	1	1		2	2	
		(ii)	I	75 [p]		1		1		
			II	Substitution: $\frac{75 \text{ ecf}}{2.50}$ (1) Cost of 1 unit = 30 [p]	1	1		2	2	
	(b)			To reduce use of fossil fuels / to reduce {greenhouse gas / carbon / CO ₂ } emissions / to prevent power cuts / to reduce {carbon footprint / climate change / global warming / greenhouse effect} Accept reduce pollution / to help the environment		1		1		
	(c)			Indicative content: The earth wire protects users of electricity. The earth wire provides a low resistance path to earth if a fault causes a metal cased appliance to become live. A fuse is used to protect circuits from overheating. If there is a short circuit causing too much current, the fuse melts. The rccb circuit breakers protect users of electricity. The rccb switches off the circuit if there is a difference between the currents in the live and neutral wires.	6			6		